



# Uncertainty and short- long-term dynamics of ecosystem services under future urbanization in Shenzhen

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## Abstract

With rapid global urbanization, the degradation of ecosystem services (ES) poses a serious threat to regional sustainability, while their inherently complex dynamics and uncertainties further complicate effective decision-making in ecosystem management. Therefore, accurate quantification of ES and their temporal changes is essential for sustainable ecosystem management. However, current ES management is constrained by limited understanding of their long- and short-term spatiotemporal variability across continuous time scales. In this study, the coefficient of variation was employed to quantify ES uncertainty based on the multi-scenario simulations of three key ES from 2022 to 2032, performed annually. Subsequently, under the ideal scenario, long-term trends of the three ES were analyzed using Theil-Sen slope estimation and the Mann–Kendall test. Finally, short-term fluctuations of ES were examined through hotspot analysis and frequency statistics, while key driving factors were identified using the GeoDetector model. The results indicate that: (1) regions with low carbon storage supply exhibited high uncertainty; (2) habitat quality showed the widest extent of extreme uncertainty, whereas water yield remained relatively stable; (3) among the three ES, water yield and habitat quality were characterized by improvements at high levels, whereas carbon storage was dominated by degradation at low levels; (4) short-term fluctuations of the three ES were concentrated along forested edges, valley tracts, and forest roads, primarily driven by land-use intensity, slope, and elevation. This study helps bridge the gap between static snapshot evaluations of ES and their intrinsic dynamics, insights essential for preventing policy miscalculations and enabling flexible adjustment of priority ecological areas in response to dynamic ecosystem changes.

**Keywords** Urbanization · Ecosystem services · Trend analysis · Short-term fluctuations · Drive detection

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# 1 Introduction

Ecosystem services (ES) refer to the essential services and functions provided by ecosystems that support human survival and development (Costanza et al., 1997; Gret-Regamey et al., 2013; Wang et al., 2025). Since ES degradation poses a serious threat to regional sustainability (Jiang et al., 2021; Pereira et al., 2024; Zhang et al., 2024b), planning based on accurate quantification of ES is considered the most effective long-term strategy to mitigate such impacts (Larigauderie & Mooney, 2010; Zhang et al., 2022). Moreover, the inherent complexity and uncertainty of ecosystems further complicate the task of making informed decisions in ecosystem management (Chen et al., 2025; Flanagan et al., 2021). Ecosystems also undergo short-term changes driven by natural conditions and management practices. It is therefore vital to rigorously assess the uncertainty and short-term fluctuations of ES and incorporate these dynamics into future adaptive management frameworks.

Recently, assessing and predicting future ES under urbanization has become a re-search hotspot, yielding extensive findings (Zhang & Li, 2023; Zhao et al., 2022). Most of these studies have examined the spatiotemporal characteristics of ES across different periods to explore their synergistic and trade-offs, supply–demand matching, service flow, urban ecological resilience, and ecological risk. For instance, Liu et al. (2022) analyzed the spatiotemporal evolution of water supply and demand and assessed the risk of their mismatch under multiple scenarios in Yulin, China. Chi et al. (2024) estimated ES flows in the Yangtze River Basin using a gravity model and proposed a standard for horizontal ecological compensation. Li et al. (2025) conducted a long-term ecological risk assessment based on ES to develop a comprehensive ecological zoning framework for riparian buffer zones. Overall, such research frameworks help reveal the interrelationships among different ES and provide valuable insights for their application in ecological management. However, the prevailing approach to ES evaluation relies on static snapshots of availability, which identify regions with high or low supply (Ding et al., 2023; Song et al., 2022; Wang et al., 2024b). Given the inherent uncertainty of ES, such snapshot assessments may quickly lose accuracy, resulting in a disconnect between actual supply and expectations. Consequently, reliance solely on snapshot results may cause managers to draw misleading inferences about ES availability (Hooftman et al., 2022; Knoke et al., 2024; Ndong et al., 2020), thereby hindering effective ecological management. Furthermore, targeted strategies are needed to address mismatches between ES supply and its development potential (Ma et al., 2024).

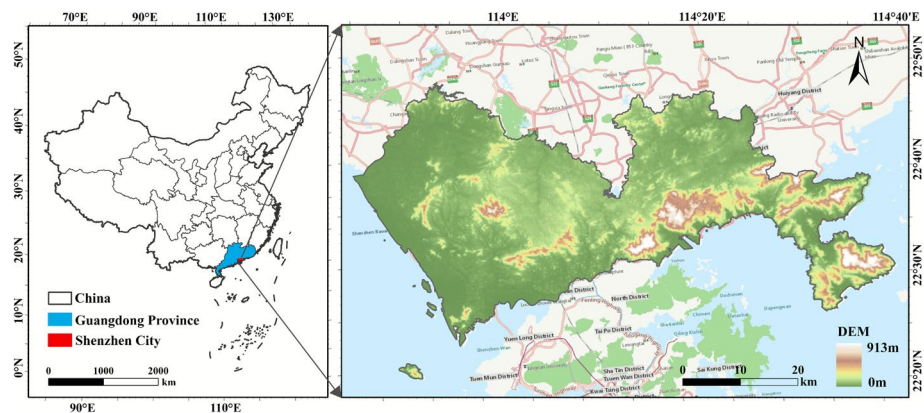
Although some studies have measured ES uncertainty using standard deviation coefficient of variation (Gret-Regamey et al., 2013), these approaches overlook the direction of change and cannot reveal whether ecosystems are improving or degrading. Trend analysis of continuous ES can address these limitations and offer insights for future ecosystem protection, restoration, and management. Currently, some studies conduct trend analysis using multi-year averages of ES at the regional scale (Hou et al., 2023; Ma et al., 2021). However, such approaches remain limited in capturing spatial heterogeneity, thereby constraining precise ecological management. Although a few studies have applied pixel-scale trend analysis to identify long-term patterns of past ES changes (Li et al., 2021; Liu et al., 2024), comprehensive examinations of future scenarios remain lacking. Moreover, short-term fluctuations of ES are complex may interfere with the identification of ecological priority areas for protection (Rau et al., 2018; Snäll et al., 2021), highlighting the urgent need for further investigation.

As one of the megacities in the Guangdong-Hong Kong-Macao Greater Bay Area, Shenzhen has experienced rapid urbanization that has caused severe disturbances to its ecosystems. In this context, clarifying the uncertainty and both the long- and short-term dynamics of ES in Shenzhen is crucial for formulating effective ecosystem management policies. The objectives of this study are: **(1)** to quantify ES uncertainty based on annual simulations for 2022–2032; **(2)** to calculate the long-term trends of ES under the optimal scenario and identify specific ecological conditions; **(3)** to reveal the short-term fluctuation of ES using hotspots analysis and frequency statistics; and **(4)** to explore the main drivers of short-term fluctuations through GeoDetector model. This study enhances the understanding of ES across continuous time scales, particularly regarding uncertainty and short-term fluctuations, which is essential for preventing policy miscalculations and supporting the development of flexible and adaptive ecological management strategies.

## 2 Study area and data sources

### 2.1 Study area

Shenzhen is located on the eastern coast of the Pearl River Estuary in Guangdong Province, China (Fig. 1), covering a total area of 1997.47 km<sup>2</sup>. Shenzhen has a southern subtropical maritime monsoon climate characterized by long, hot summers, short winters, abundant precipitation, and a mean annual temperature of 22.3 °C. The topography is higher in the east and lower in the west, with low mountains, hills, and plateaus accounting for 62.8% of the total land area. The city has a well-developed hydrological system. In 2020, Shenzhen's GDP exceeded 2.7 trillion yuan, ranking among the top five in Asia, while its per capita GDP exceeded 150,000 yuan, ranking first among Chinese cities. With a resident population of 13.03 million, Shenzhen became the first fully urbanized city in China. Its rapid urbanization and intensive human activities have exerted substantial pressure on ecosystems, making it a representative case for studying the dynamics and uncertainty of ecosystem services.



**Fig. 1** Location map of Shenzhen City

## 2.2 Data sources and preprocessing

The datasets used in this study include land-use data, natural and socio-economic data for ES evaluation, and projection datasets for analyzing the drivers of short-term ES fluctuations. Detailed descriptions and sources of the datasets are provided in Table 1. Considering the characteristics of the study area, land-use types were reclassified into six categories: cropland, forest, built-up land, water, grassland, and wetland. All land-use driver datasets were resampled to a spatial resolution of 30 m in ArcGIS 10.8 and spatially aligned at the pixel level. Future projection data were subsequently preprocessing through projection transformation, clipping, and raster calculations.

## 3 Methodology

The technical framework of this study (Fig. 2) consists of four main steps. (1) Multi-scenario annual simulations of the three major ES in Shenzhen for 2022–2032 were conducted using the PLUS-InVEST model, providing the data foundation for this study. (2) Under the optimal scenario, ES uncertainty was quantified and classified using the coefficient of variation (CV). (3) To further clarify the ecological implications of ES uncertainty, long-term ES trends were analyzed at the pixel scale using Theil-Sen slope estimation and the Mann–Kendall test. The results of the trend analysis, combined with ES supply classification, were used to identify specific ecological conditions. (4) Short-term ES fluctuations were investigated by calculating hotspot frequency and interannual fluctuation ranges. Static and dynamic indicators were then applied in the GeoDetector model to identify the drivers of these fluctuations.

**Table 1** Parameters of data used and their sources

| Category                  | Product         | Source                                                                                                                                                                          |
|---------------------------|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LUCC                      | CLCD            | The 30 m annual land cover datasets and its dynamics in China from 1985 to 2022 ( <a href="https://doi.org/10.5281/zenodo.4417809">https://doi.org/10.5281/zenodo.4417809</a> ) |
| Natural Drivers           | Water           | OpenStreetMap ( <a href="https://www.openstreetmap.org/">https://www.openstreetmap.org/</a> )                                                                                   |
|                           | DEM             | Geospatial Data Cloud (China) ( <a href="https://www.gscloud.cn/">https://www.gscloud.cn/</a> )                                                                                 |
|                           | Slope           | Calculated based on DEM                                                                                                                                                         |
|                           | Aspect          |                                                                                                                                                                                 |
|                           | NDVI            | Google Earth Engine ( <a href="https://earthengine.google.com/">https://earthengine.google.com/</a> )                                                                           |
|                           | Precipitation   | Resources and Environmental Science Data Center ( <a href="https://www.resdc.cn">https://www.resdc.cn</a> )                                                                     |
|                           | Temperatures    |                                                                                                                                                                                 |
|                           | GPP<br>NPP      |                                                                                                                                                                                 |
| Socio-economic drivers    | Road            | OpenStreetMap ( <a href="https://www.openstreetmap.org/">https://www.openstreetmap.org/</a> )                                                                                   |
|                           | POI             | Amaps Open Platform ( <a href="https://lbs.amap.com/">https://lbs.amap.com/</a> )                                                                                               |
|                           | Population      | WorldPop ( <a href="https://www.worldpop.org">https://www.worldpop.org</a> )                                                                                                    |
|                           | GDP<br>NTL      | Resources and Environmental Science Data Center ( <a href="https://www.resdc.cn">https://www.resdc.cn</a> )                                                                     |
| Future prediction dataset | Population      | Global 1 KM-Grid Population Distributions from 2020 to 2100 ( <a href="https://doi.org/10.6084/m9.figshare.19608594.v2">https://doi.org/10.6084/m9.figshare.19608594.v2</a> )   |
|                           | Climate factors | National Tibetan Plateau Data Center ( <a href="https://data.tpdac.ac.cn/">https://data.tpdac.ac.cn/</a> )                                                                      |

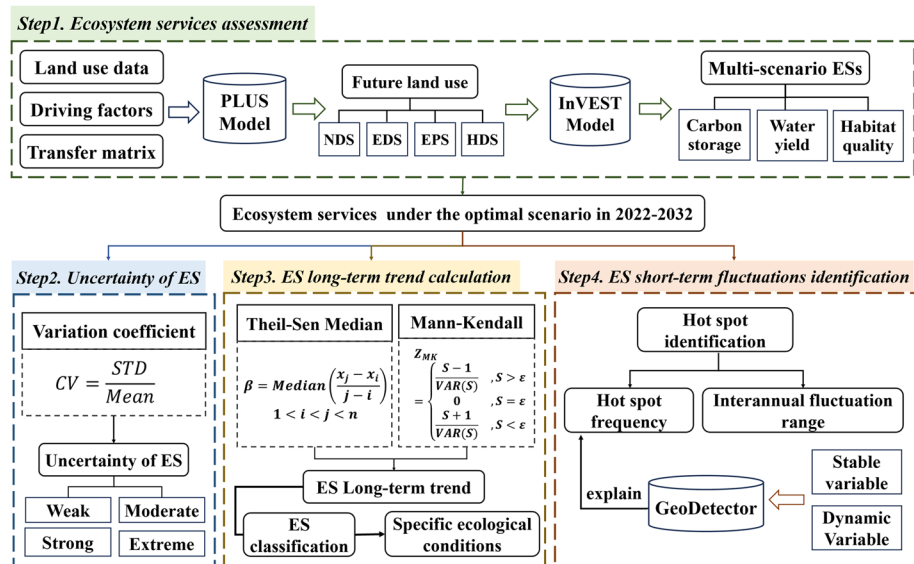


Fig. 2 Technological route map of this study

### 3.1 Future land-use & ES modeling pipeline

#### 3.1.1 Multi-scenario design

Drawing on existing studies (Baier et al., 2009; Sun & Liang, 2021; Wu & Wang, 2025) and Shenzhen's current development context, four scenarios-natural development, ecological protection, economic development, and harmonious development-were designed using the Markov model. Specifically, the design followed established principles and approaches for scenario setting as outlined in previous studies. Furthermore, the Shenzhen Master Plan for Territorial Space (2020–2035) was incorporated to strengthen the policy relevance of the scenario design. In Markov models, land-use transition probabilities at the next time step are primarily determined by the current state and neighborhood effects, and this approach has been widely applied in multi-scenario studies.

In the natural development scenario, the land-use transfer matrix was derived from the 2012–2022 Markov process and remained unchanged for 2022–2032. Following previous research on adjusting Markov transition probability adjustment (Shi et al., 2023; Yang et al., 2023), the land-use transfer matrix for 2022–2032 was modified to project land-use demand under the other three scenarios. In the economic development scenario, priority was given to urban expansion by increasing the probability of converting ecological land to built-up land and reducing the probability of converting built-up land to other land types (except cropland). In the ecological protection scenario, the expansion of cropland and built-up land was strictly controlled, giving priority to ecological benefits. Net primary productivity (NPP) values of forest and cropland were extracted, and areas with medium to high values were designated as restricted zones. This scenario allowed forest areas to expand over time while reducing, to varying degrees, the probability of converting cropland and ecological land to built-up land. In the harmonious development scenario, both economic growth and

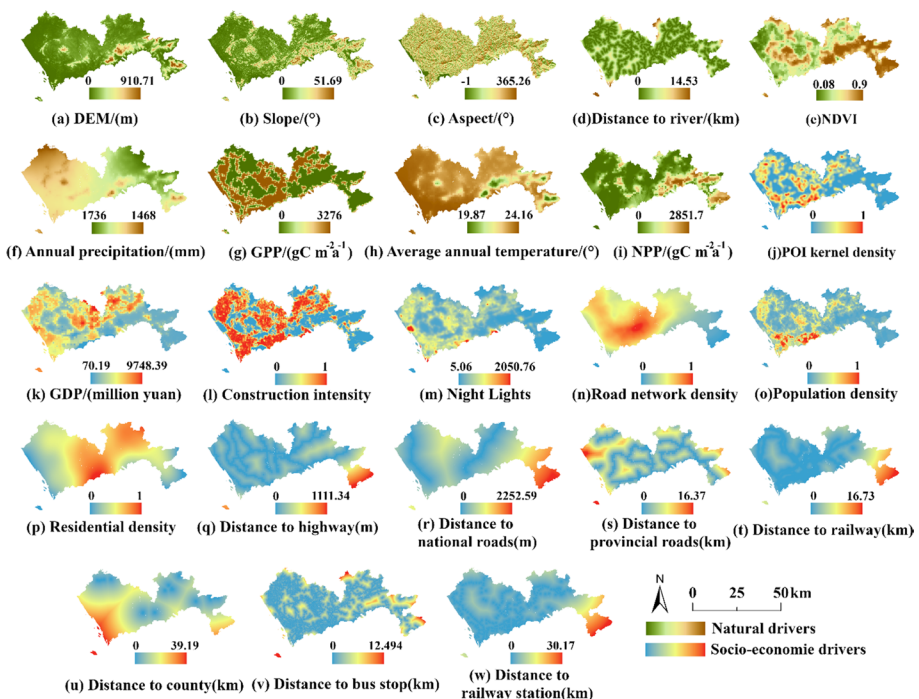
ecological protection were pursued. The probability of converting other land to forest was increased, while the probability of converting cropland and grassland to built-up land was moderately raised, with high-NPP grassland and forest areas designated as restrictive zones.

### 3.1.2 Land-use simulation

Projecting ES from future land-use patterns is an effective approach for assessing ES dynamics. As an improved cellular automata (CA) model, the PLUS model has been widely applied to identify the drivers of land expansion and to simulate land-use changes at the patch level. Detailed formulations of the PLUS model are available in Liang et al. (2021). A total of 23 drivers—9 natural and 14 socio-economic factors (Fig. 3)—were selected to model land-use changes under the four scenarios using the PLUS model (Gao et al., 2022; Jian et al., 2024; Zhang et al., 2024a). To validate simulation accuracy, the predicted land-use maps were compared with actual land-use data for 2022. Validation results indicated a Kappa coefficient of 0.82, an FOM coefficient of 0.102, and an overall accuracy of 86%, demonstrating that the PLUS model provides sufficient accuracy to support ES simulations with the InVEST model.

## 3.2 ES modeling

In this study, the InVEST model was employed to assess three key ES under future land-use scenarios. Future multi-scenario land-use data generated by the PLUS model served



**Fig. 3** Spatial driving factors of land use change

as the primary input for the InVEST model, enabling the assessment of how different land management decisions influence ES provision through land-cover change. The InVEST model provides a widely used approach for quantifying ES functions and offers a scientific basis for evaluating the impacts of human activities by simulating ES changes under different land-cover scenarios (García-Ontiyuelo et al., 2024). Urbanization reduces regional water-holding capacity through the expansion of impervious surfaces, leads to biodiversity decline, and affects the global carbon cycle and climate change (Fryer & Williams, 2021; Rosa et al., 2020). In this study, habitat quality, water yield, and carbon storage modules were applied, with their specific formulations presented in Table 2.

3.3 Metrics for ES dynamics

3.3.1 Uncertainty quantification

The CV is a statistic measure of variability and is commonly used to assess the temporal stability of spatial data. In this study, CV was applied to quantify ES uncertainty, defined as the ratio of the standard deviation to the mean, thereby eliminating the influence of unit differences. Following Yang et al. (2024), ES uncertainty was classified into four levels: weak (CV between 0 and 15%), moderate (15–36%), strong (36–100%), and extreme (greater than 100%). The specific formula for CV is expressed as follows:

$$CV = \frac{\sigma}{\mu} \times 100\%$$
 (1)

where  $\sigma$  is the standard deviation of the time series data and  $\mu$  is its mean.

Table 2 Required parameters and calculations for the InVEST model

| Service type    | formulation                                                              | Variable description                                                                                                                                                                                                                                                                                                                 |
|-----------------|--------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Habitat quality | $HQ_i = H_i \left[ 1 - \left( \frac{D_i^z}{D_i^z + k^z} \right) \right]$ | $HQ_i$ ranges from 0 to 1 at the pixel , with higher scores representing higher habitat quality and assuming a greater species richness. $H_i$ is the habitat suitability of the pixel . The $z$ is the scaling parameters, and $K$ is the half-saturation constant. $D_i$ is the total threat level of land use and land cover type |
| Water yield     | $WY_x = P_x \left( 1 - \frac{AET_x}{P_x} \right)$                        | $WY_x$ is the water supply of pixel $x$ , $AET_x$ is the actual vapor transpiration, and $P_x$ is the precipitation                                                                                                                                                                                                                  |
| Carbon storage  | $CS_x = C_{above} + C_{below} + C_{soil} + C_{dead}$                     | $CS_x$ is the total carbon storage of pixel $x$ , $C_{above}$ is the aboveground carbon density, $C_{soil}$ is the soil organic carbon density and $C_{dead}$ is the dead organic carbon density                                                                                                                                     |



### 3.3.2 Long-term trend analysis

The Theil-Sen estimator was applied to analyze pixel-scale trends of the three ES from 2022 to 2032, and trend significance was evaluated using the Mann–Kendall test. The Theil-Sen estimator is a robust non-parametric statistical method for trend calculation, capable of minimizing the influence of missing data and reducing the impact of outliers (Feng & Zeng, 2022). In addition, the Mann–Kendall test is a non-parametric procedure that does not assume normality and is insensitive to missing values and outliers. It was employed here as a complement to the Theil-Sen estimator to test the significance of ES time-series trends.

Long-term ES trends were classified into five categories: significant improvement, slight improvement, stable, slight degradation, and significant degradation, based on  $\beta$  and  $Z$  values at the 95% confidence level. Trend analysis results were overlaid with the 2022 spatial distribution of ES to identify five ecological conditions: improvement with high ES, improvement with low ES, remaining stable, degradation with high ES, and degradation with low ES.

### 3.3.3 Short-term fluctuation metrics

Short-term fluctuations of ES were assessed based on variations in hotspot frequency over the 10-year period (2022–2032). The analysis was conducted in three steps, as follows:

i. Hotspot identification (percentile)

The percentile method is a widely used standardized threshold approach for hotspot identification in spatial ecology (Fu et al., 2025; Thom et al., 2017; Wang et al., 2024c; Zhao et al., 2025). It efficiently identifies annual hotspots from raster data, rendering it suitable for long-term time-series comparisons. In this study, it provided the basis for quantifying hotspot persistence and interannual fluctuations. Specifically, the top 50% of pixels ranked by ES values in a given year were defined as hotspots, while the bottom 50% were designated as coldspots, allowing the identification of hotspot regions for each ES across all 11 periods.

ii. Hotspot persistence (frequency)

To capture the temporal variability of ES hotspots, interannual fluctuations can be conceptualized as stochastic processes. When a spatial unit consistently falls within the high-value range across multiple observations, its probability of high provision exceeds random expectations, indicating statistical stability; otherwise, it exhibits volatility. Accordingly, the number of times a pixel meets the hotspot threshold can be used to measure its temporal consistency. In this study, annual hotspot and coldspot maps were binarized, and 11 such images were stacked to calculate the frequency of hotspot occurrence for each pixel. Specifically, a frequency of 11 indicates that a pixel remained a hotspot throughout the period, a frequency of 0 indicates that it was consistently a coldspot, and frequencies between 1 and 10 reflect varying degrees of fluctuation. Thus, the degree of ES fluctuation and its spatial heterogeneity were derived.

iii. Interannual fluctuation range (IFR)

The interannual fluctuation range (IFR) was calculated as the overlap area of ES hotspots between adjacent years (Wang et al., 2024a), where larger values indicate more significant interannual spatial displacement of ES.



$$IFR_i = S - S_i, \quad 0 < i < 11 \quad (2)$$

where  $IFR_i$  denotes the interannual fluctuation range in year  $i$ ,  $S_i$  denotes the intersection area of ES hotspots in years  $i - 1$  and year  $i$ , i.e., the overlap area of ES hotspots in adjacent years, and  $S$  denotes the hotspot area in year  $i$ .

### 3.4 Drivers of short-term fluctuations

#### 3.4.1 Variable system and preprocessing

The GeoDetector model was employed to identify the drivers of short-term ES fluctuations, owing to its advantages of avoiding linearity assumptions and remaining unaffected by multicollinearity among independent variables (Wang & Xu, 2017). In this framework, the hotspot frequency of each pixel during 2022–2032, representing the cumulative number of times a pixel was classified as a hotspot, was taken as the dependent variable. This indicator not only reflects fluctuation magnitude but also preserves spatial location information, thereby aligning with the spatial stratified heterogeneity principle of GeoDetector model.

Nine potential influencing factors were selected as independent variables, considering data availability, ecological mechanisms, and prior research findings. These were classified into **static variables** (e.g., slope and terrain) and **dynamic variables** (e.g., temperature, precipitation, evapotranspiration, population density, and land use intensity). Static factors capture inherent topographic characteristics, which remain stable on an interannual scale, whereas dynamic factors vary annually and represent external forcing on ecosystem processes. To reduce noise and emphasize long-term tendencies, the mean values of the dynamic factors over the 10-year simulation period were used. Land use intensity was assigned values following Yang et al. (2021), as detailed in Table 3.

#### 3.4.2 Factor and interaction detection

The GeoDetector framework was employed to analyze both single-factor and interactive effects on short-term ES fluctuations. The factor detection module quantifies the explanatory power of individual variables through  $q$ -values ranging from 0 to 1, with higher values indicating stronger associations between the factor and the dependent variable. This allows for identifying the dominant drivers among natural and anthropogenic factors.

In addition, the interaction detection module evaluates whether the combined effects of two variables enhance or weaken their individual explanatory powers. By comparing the joint  $q$ -value of two factors with their respective single-factor  $q$ -values, interactions can be classified as bi-factor enhancement, non-linear enhancement, independence, or weakening. This approach is essential for uncovering potential synergistic drivers, such as the interaction between land use intensity and topographic or climatic factors.

**Table 3** Assignment of land use intensity

| Land use type   | Forest, grassland, water and wetland | Cropland | Built-up land |
|-----------------|--------------------------------------|----------|---------------|
| Intensity class | weak                                 | medium   | strong        |
| Rank assignment | 1                                    | 2        | 3             |

### 3.5 Robustness & method validation

Emerging Hot Spot Analysis (EHA) identifies statistically significant spatiotemporal trends by categorizing areas into classes such as persistent and intensifying hotspots. This approach provides a dynamic perspective on the evolution of hotspot patterns over time. To validate our approach, hotspots identified by EHA were compared with those derived from the percentile method used in this study. The results demonstrated strong agreement, especially in persistent hotspots and areas of fluctuation, thereby confirming the robustness of the percentile-based method.

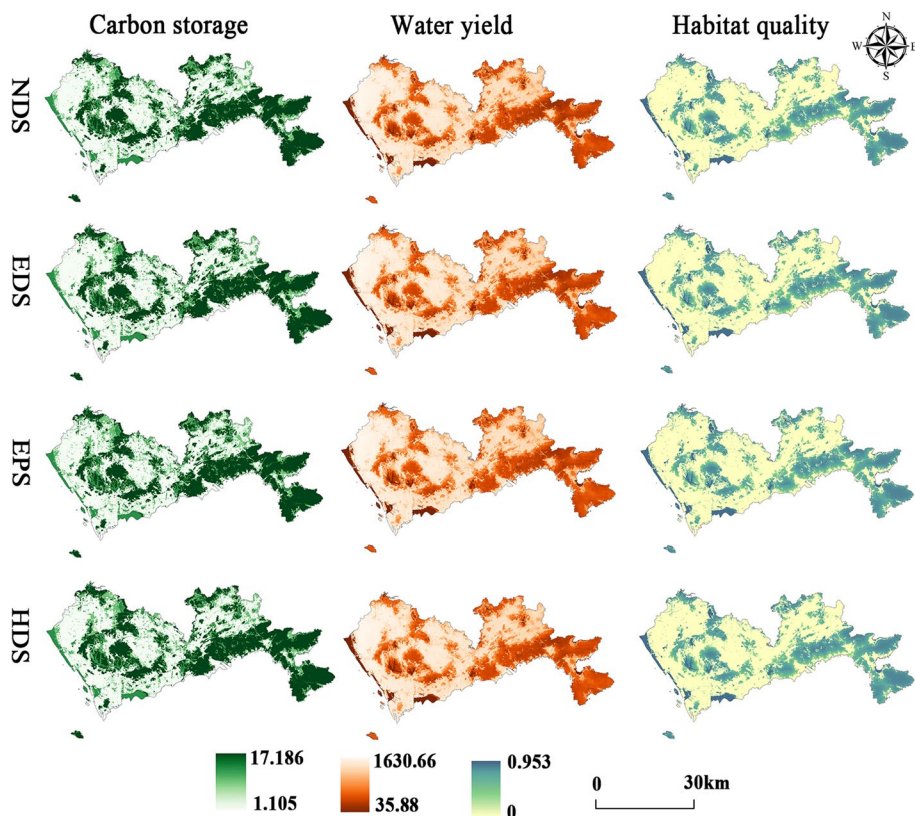
Although EHA is an effective tool for hotspot identification, the percentile-based approach offers distinct advantages for the present research context. The percentile method allows efficient delineation of annual hotspot areas directly from raster data and facilitates pixel-scale hotspot frequency calculations without extensive preprocessing. In contrast, EHA requires the construction of a space–time cube from multi-year ES data, a computationally intensive process that often necessitates down sampling high-resolution ES outputs to coarser resolutions to ensure feasibility. Such resampling may compromise the precision of ecological assessments. Furthermore, EHA consolidates all 11 periods of ES data into a single output map that categorizes spatiotemporal patterns, but does not support the identification of annual hotspot extents. Therefore, EHA cannot be applied to the fourth step of our methodology—the calculation of the interannual fluctuation range—which requires year-to-year spatial shifts in hotspot distributions.

## 4 Results

### 4.1 The assessment of ES under multi-scenario simulation

Based on the 2032 multi-scenario land-use simulations, the InVEST model was employed to assess water yield, carbon storage, and habitat quality in Shenzhen under different scenarios. Spatially, carbon storage and habitat quality exhibited a polarized pattern, with higher values in the east and lower values in the west, where high values were concentrated in areas of dense vegetation cover (Fig. 4). These regions were typically forested, characterized by high carbon sequestration capacity, relatively low human disturbance, and well-preserved habitats. Water yield exhibited the opposite pattern, mainly because the western region was dominated by urban built-up areas with a high proportion of impervious surfaces and low evapotranspiration. The east–west polarization of ES was highly consistent with Shenzhen's three-step topography: the mountainous east naturally limited urban expansion, whereas the western plain has long been the primary development zone, highlighting the dominant influence of topography on ES distribution. Moreover, land-use dynamics were strongly associated with variations in the three ES, with built-up land mainly contributing to water yield and forest primarily supporting carbon storage (Fig. 5).

Although the spatial distribution of ES appeared similar across the four scenarios, the responses of the three ES differed markedly. Under the natural development scenario (NDS), habitat quality was lowest (0.4915), with degraded regions concentrated on hilltops and near reservoirs. As hilltops often function as biodiversity refuges, their habitat degradation highlights deficiencies in current conservation policy. In the economic development



**Fig. 4** Spatial patterns of ES in 2032 under different scenarios

scenario (EDS), water yield reached its highest level, increasing by  $1.7 \times 10^7$  Mg relative to 2022. This was attributable to a significant expansion of built-up land, resulting in a persistently high proportion of impervious surfaces and reduced evapotranspiration. However, the reduced evapotranspiration poses potential risks by exacerbating the urban heat island effect, contradicting the objective of sponge city construction aimed at enhancing rainwater infiltration. Overall, the ES response under this scenario was suboptimal: habitat quality declined to the second-lowest level after NDS, and carbon storage reached the lowest value ( $1.58 \times 10^7$  Mg). In the ecological protection scenario (EPS), habitat quality remained highest, maintaining 2022 levels and preventing further deterioration. Spatially, both degraded and improved habitat patches were observed relative to 2022. Degraded areas were mainly distributed along feeder roads, whereas improvements dominated in low-elevation forests near reservoirs. This indicates that habitat fragmentation along road edges can exert barrier effect even under ecological priority policies. Transforming small side roads into greenways represents a feasible option, as exemplified by Shenzhen's Yanlong Road renovation project, which replaced concrete drains with vegetated swales. In the harmonious development scenario (HDS), carbon storage was optimal, increasing by  $3.66 \times 10^3$  Mg relative to 2022. Habitat quality showed the least degradation, while water yield increased by  $2 \times 10^6$  mm.

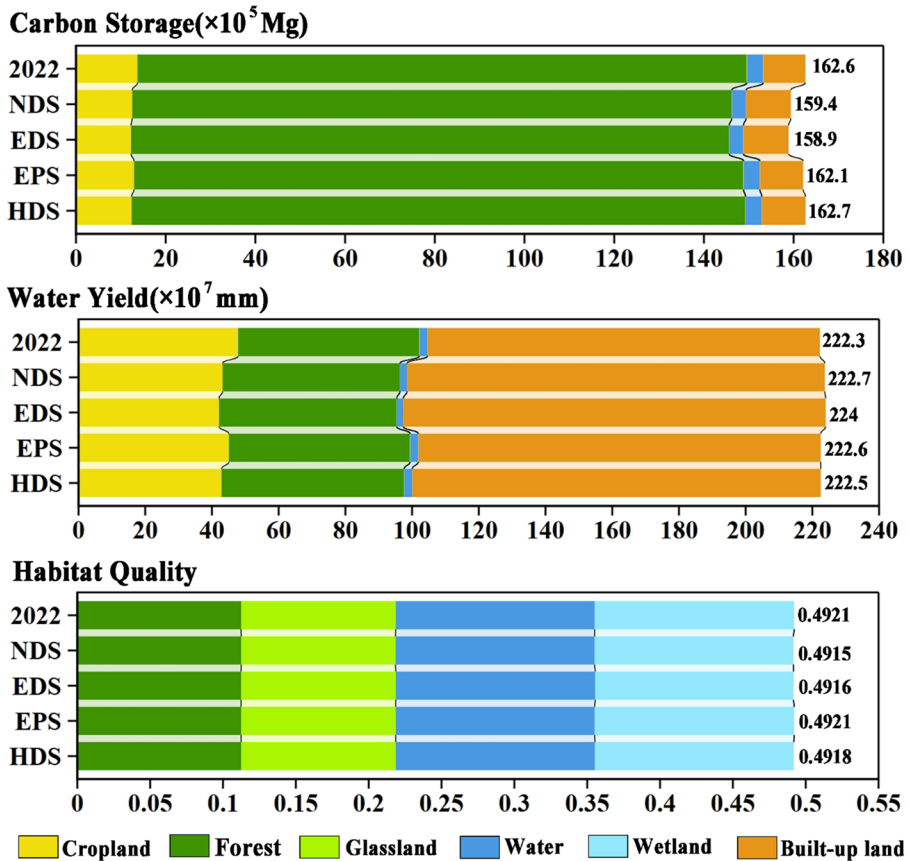
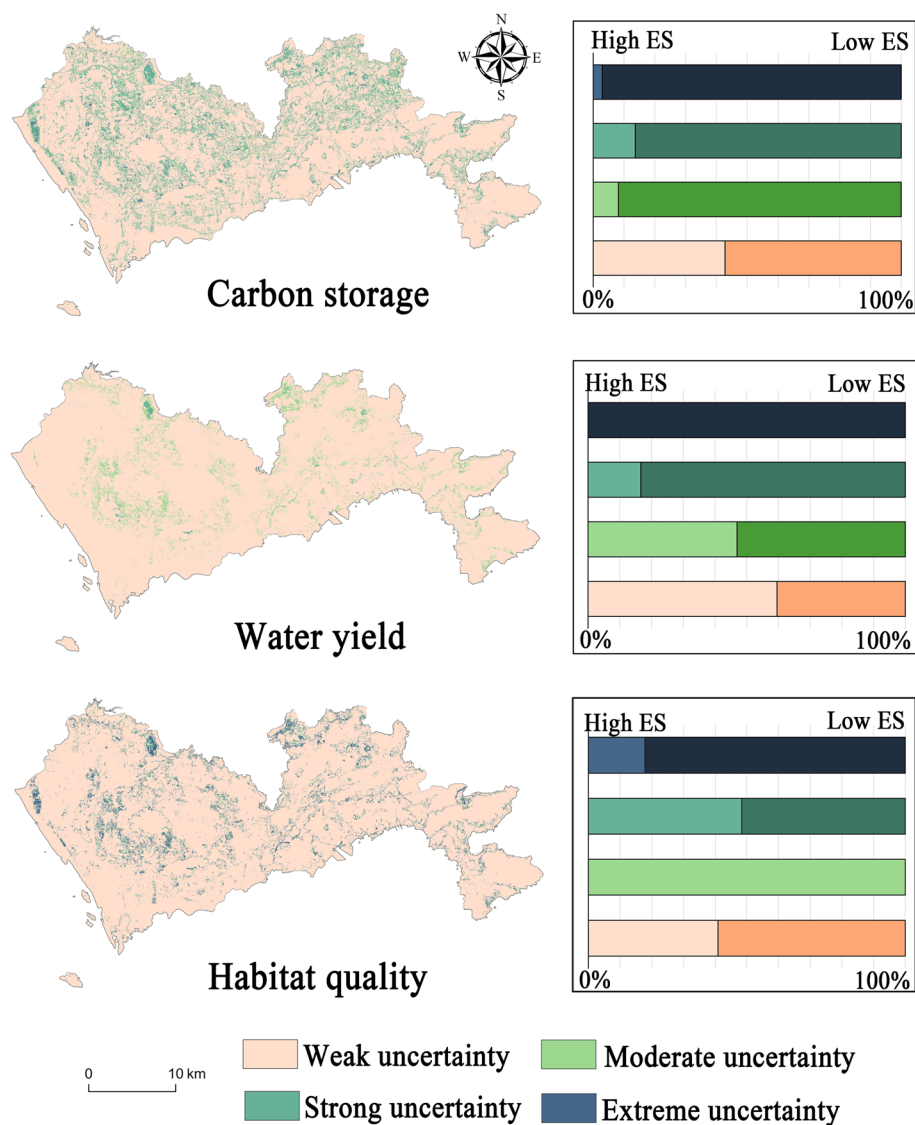


Fig. 5 Temporal changes of ES supply under different scenarios

Thus, the overall ES response in this scenario was favorable, indicating that regional sustainability can be achieved.

#### 4.2 Uncertainty of ES under the optimal scenario

The analysis of ES uncertainty under the optimal scenario revealed distinct patterns across the three services (Fig. 6). Overall, carbon storage was characterized mainly by strong uncertainty in addition to weak uncertainty, habitat quality by extreme uncertainty, and water yield by moderate uncertainty. Spatially, strong carbon storage uncertainty was concentrated in strips along major transportation corridors, whereas extreme uncertainty appeared as scattered patches in foothill areas, suggesting that infrastructure development associated with urbanization may increase the instability of carbon storage capacity. For habitat quality, strong uncertainty was concentrated in the southeastern coastal region and along rivers in the west. Water yield exhibited moderate uncertainty, especially around Xili Reservoir and Tiegang Reservoir, indicating potential influences of hydrologic management or interannual precipitation variability on water supply stability.



**Fig. 6** Spatial distribution of uncertainty in ES

When ES supply levels were considered, areas with low carbon storage exhibited higher uncertainty, with strong uncertainty covering 213 km<sup>2</sup> (Table 4). These regions were particularly vulnerable to climate and land-use change due to sparse vegetation cover and unstable soil organic carbon reservoirs, while extreme uncertainty was almost entirely confined to low-supply areas (16 km<sup>2</sup>). Habitat quality had the largest area of extreme uncertainty (92 km<sup>2</sup>), also predominantly located in low-supply regions. Water yield was relatively stable, dominated by moderate and weak uncertainty, and extended across both high- and low-supply areas. This pattern reflects the buffering capacity of Shenzhen's water supply

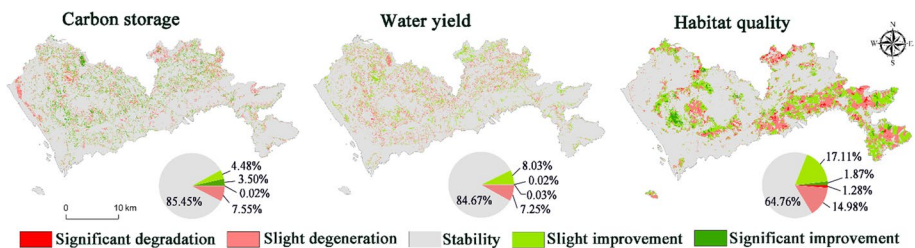
**Table 4** Area of uncertainty statistics for ES (km<sup>2</sup>)

| Uncertainty | Carbon storage |            | Water yield |            | Habitat quality |            |
|-------------|----------------|------------|-------------|------------|-----------------|------------|
|             | High supply    | Low supply | High supply | Low supply | High supply     | Low supply |
| Weak        | 673.109        | 900.023    | 1057.942    | 719.187    | 710.852         | 1026.401   |
| Moderate    | 4.359          | 48.764     | 48.311      | 54.566     | 0.169           | 0.000      |
| Strong      | 33.816         | 213.440    | 1.657       | 8.315      | 29.479          | 31.416     |
| Extreme     | 0.4851         | 15.9948    | 0           | 0.0117     | 16.389          | 75.2832    |

system, although the potential impacts of extreme weather events on reservoir basins remain a concern. Overall, ES uncertainty reflects not only the likelihood of degradation but also the ecological potential for optimization, and the subsequent trend analysis builds on this to determine the specific ecological status of each region.

### 4.3 Long-term trends of ES under the optimal scenario

To clarify the directions of ES change underlying their uncertainty, trends of the three ES from 2022 to 2032 were calculated under the optimal scenario. The results (Fig. 7) indicated that all three ES exhibited a coexistence of localized improvement and degradation over the next decade. For carbon storage and water yield, the areas of improvement and degradation were approximately equal, whereas the area of improved habitat quality exceeded the degraded area by 49.86 km<sup>2</sup>. Degraded areas of carbon storage were mainly distributed in narrow strips along major roads, such as Shenzhen Outer Ring Highway and the Eastern Transit Highway. This was likely due to continued road construction cutting through natural habitats, leading to vegetation fragmentation. In this context, constructing green bridges and replanting native tree species in deteriorated regions such as the Outer Ring Road represent practical mitigation measures. Areas of improved carbon storage were concentrated along the edges of cropland and forest, suggesting that reduced agricultural activity during urbanization facilitated secondary forest regeneration and enhanced carbon storage capacity. Trends in water yield were relatively stable, with slightly degradation mainly occurring at the borders of built-up land and cropland, and slightly improvements concentrated near roads and industrial parks. This was attributed to increased impervious surfaces reducing rainwater infiltration. In the long term, sponge city construction provides a more sustainable solution. Habitat quality trends were most pronounced in woodlands in the southeastern and west-central areas of Shenzhen, with degradation dominating around reservoirs in the northeast and improvement prevailing in the west-central region. These patterns may

**Fig. 7** Trends in ES under HDS in 2022–2032

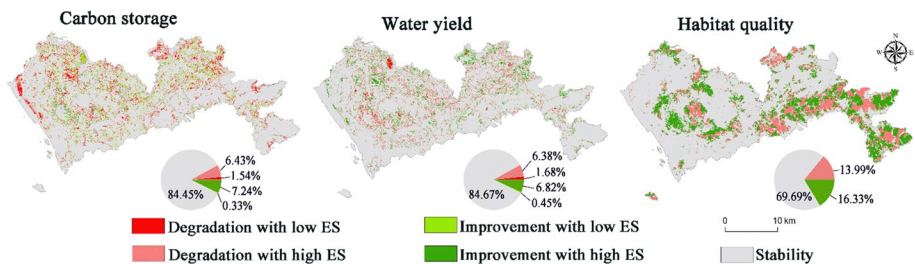


be linked to reduced vegetation adaptability caused by tourism development and climate change. Management strategies should include controlling tourist density and designating ecological protection zones in degraded areas.

Following the trend analysis, the initial state of ES distribution in 2022 was also considered. In other words, the ES trends were combined with their 2022 spatial distribution to classify ecological status into five categories (Fig. 8). Overall, water yield and habitat quality were mainly characterized by improvement with high ES and by stability. Although total carbon storage increased from 2022 to 2032, its ecological status was dominated by degradation with low ES. This indicates that policymakers should be cautious of illusory improvements in ES and instead emphasize the quality of ecological restoration. Carbon storage degradation was mainly observed in fragmented croplands, where land-use fragmentation disrupted continuous carbon sinks. These areas clustered in the west within large built-up zones such as convention centers, airports, and industrial parks, while in the east they appeared as striped patterns along forest roads. Therefore, rather than focusing solely on total carbon stock compliance, management should prioritize ecological restoration in low-ES and degraded areas. For water yield, areas of improvement with high-ES dominated, aside from stable regions, and were mainly located in woodlands, large-scale plantations, and forest roads near reservoirs. Degraded high-ES water yield areas also require attention, particularly at the junctions of cropland and built-up land. Establishing buffer strips of approximately 10 m in degraded areas is recommended to compensate for natural hydrological functions. Changes in habitat quality trends were mainly concentrated in ecological land. Notably, high-altitude mountains with favorable natural conditions and limited human disturbance still experienced degradation in areas of high habitat quality.

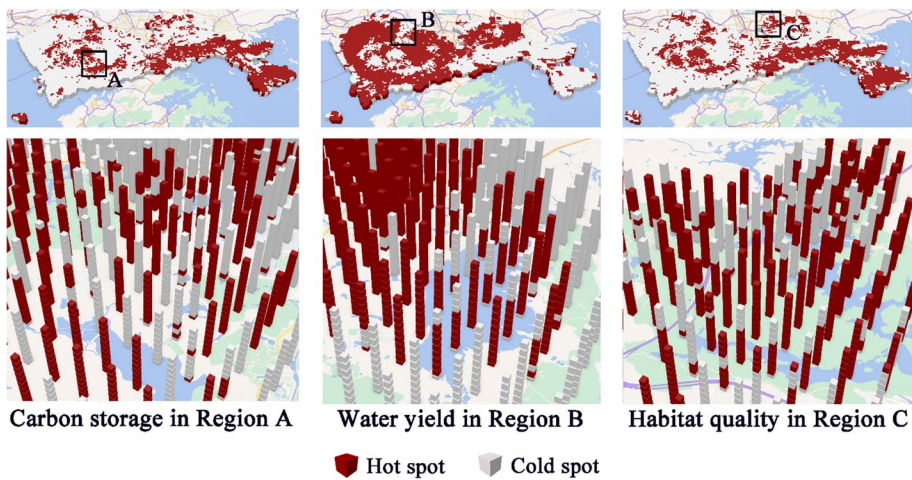
#### 4.4 Short-term fluctuations of ES under the optimal scenario

To visualize the interannual variability of ES hotspots, a space–time cube was constructed using annual hotspot data. Enlarged views of the cube for the three ES (Fig. 9) revealed that many pixels transitioned between hotspot and coldspot status across years, rather than remaining consistently classified. This spatio-temporal pattern provides direct visual evidence of the inherent volatility in ES supply. Specifically, some pixels were identified as hotspots only in the initial or final years of the study period, whereas others repeatedly switched between hotspot and coldspot status during intermediate years. Therefore, to complement this visual evidence, hotspot frequency at the pixel scale was calculated to quantify the observed volatility.



**Fig. 8** Identification of specific ecological status of Shenzhen in 2022–2032



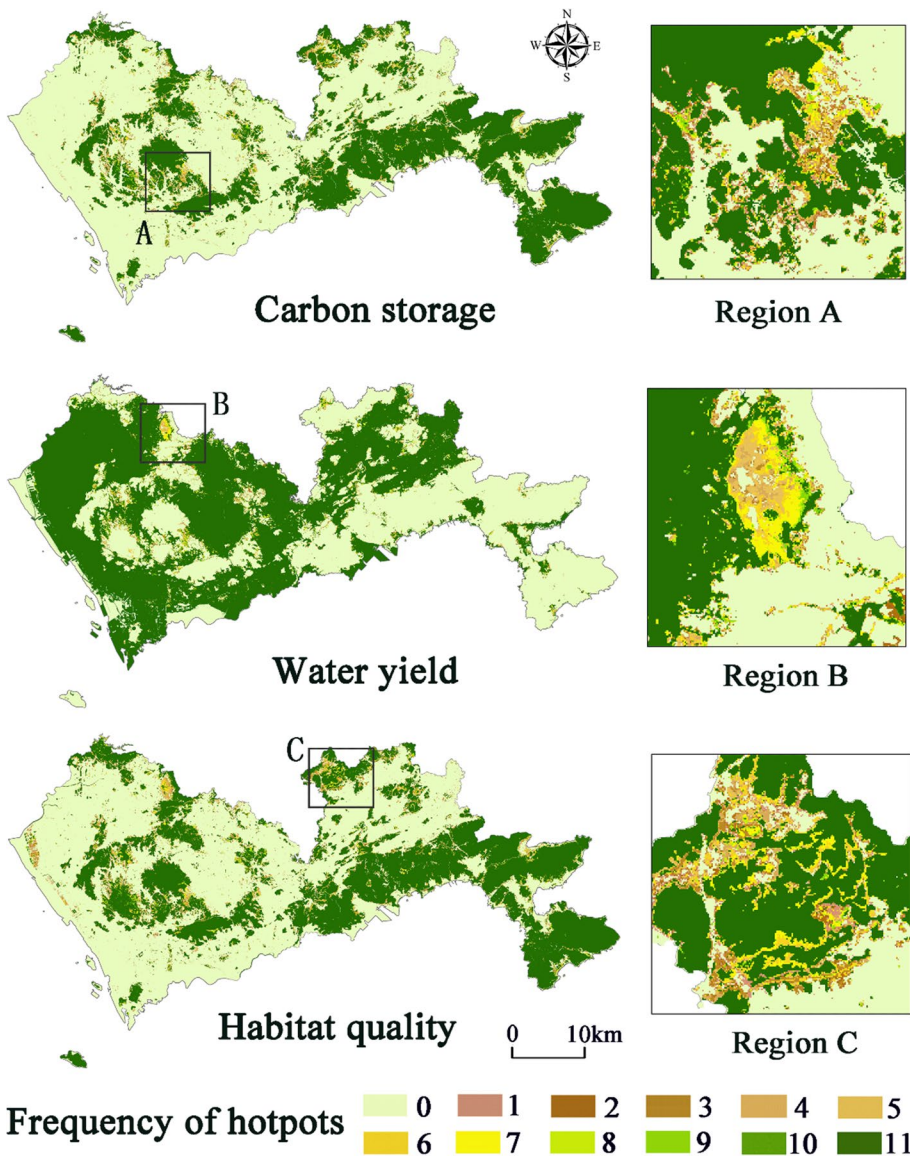


**Fig. 9** The annual hotspots of 2022–2032 under the Space–Time Cube

**Table 5** Statistics of different hotspot frequency areas in 2022–2032 (km<sup>2</sup>)

| Hotspot frequency | Carbon storage | Water yield | Habitat quality |
|-------------------|----------------|-------------|-----------------|
| 1–2               | 39.33          | 15.75       | 42.80           |
| 3–4               | 29.17          | 18.66       | 33.56           |
| 5–6               | 27.15          | 20.38       | 31.67           |
| 7–8               | 24.17          | 21.76       | 27.82           |
| 9–10              | 15.12          | 29.69       | 16.71           |
| 11                | 673.22         | 980.77      | 711.02          |

The frequency distribution of ES hotspots in Shenzhen over the next decade reflected short-term fluctuations across all three ES (Fig. 9). Constant hotspot areas (frequency = 11) for habitat quality and carbon storage were concentrated in southeastern and central-western forests. By contrast, constant hotspots for water yield were concentrated in western built-up land, showing an opposite polarization pattern. Regions with hotspot frequencies ranging from 1 to 10 represented areas of short-term ES fluctuations, with the frequency values indicating different degrees of volatility. Spatially, the fluctuation areas of the three ES exhibited overall similar distribution patterns. Low-frequency hotspots were primarily clustered along forest edges and valleys, forming distinct bands along forest roads and tunnels. Medium-frequency hotspots were primarily distributed around reservoirs, forming circular patterns. In addition, habitat quality exhibited continuous low-frequency hotspots around western convention centers and airports. Fluctuation areas of habitat quality and water yield encompassed entire reservoirs, with frequency gradually decreasing and volatility increasing from the edges toward the reservoir centers. Statistics on the area of different hotspot frequencies for the three ES showed that habitat quality had the largest total area of hotspot fluctuations, followed by carbon storage (Table 5). This order was consistent with that of persistent hotspots. This suggests that regions with larger hotspot extents were also accompanied by correspondingly larger fluctuations. Carbon storage and habitat quality were dominated by low-frequency hotspot which accounted for 50% of their total fluctuating area, whereas water yield was primarily characterized by medium- to high-frequency hotspots (Fig. 10).



**Fig. 10** Hotspot frequency of ES in Shenzhen from 2022 to 2032

After clarifying the spatial distribution pattern of short-term ES fluctuations interannual fluctuation ranges (IFR) of the three ES were measured to capture year-to-year variability. Specifically, the spatial offset of each year's hotspot relative to the previous year was calculated across the entire study area over the 10-year period (Fig. 11). The magnitude of change in hotspot overlap between neighboring years was substantially larger than the changes in hotspot extent, indicating that hotspot locations were displaced to varying degrees between years. The annual IFR trends of the three ES were broadly comparable. On average, water

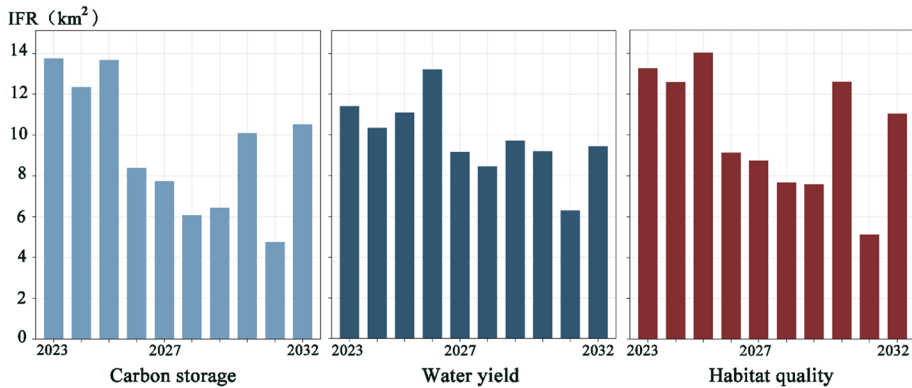


Fig. 11 Interannual fluctuation range of ES in Shenzhen in 2022–2032

yield exhibited the smallest IFR and variance, indicating greater overlap of hotspot areas between consecutive years. Habitat quality had the highest mean IFR and variance, followed by carbon storage. The IFR of habitat quality peaked at  $14 \text{ km}^2$ , showing substantial variability in hotspot locations. Carbon storage IFR reached up to  $5 \text{ km}^2$ , while habitat quality reached to  $7 \text{ km}^2$ .

These findings implies that built-up land expansion may exacerbate spatial instability in ES by altering surface permeability and disrupting vegetation continuity. Moreover, future extreme weather events may heighten the sensitivity of biological processes, leading to abrupt shifts in marginal habitats. In summary, short-term fluctuations were most pronounced in habitat quality, followed by carbon storage, while water yield showed relatively stable variations. These patterns challenge the adequacy of traditional static conservation strategies.

#### 4.5 Drivers of short-term fluctuations in ES

This study investigated the drivers of short-term fluctuations of ES through factor and interaction detection with the GeoDetector model. The single-factor results (Table 6) indicated that land use intensity was the most important driver of short-term fluctuations in ES hotspots. For carbon storage, slope exerted the strongest influence ( $q=0.542$ ), followed by elevation and annual evapotranspiration, both with  $q$ -values above 0.4. For water yield, annual evapotranspiration explained more than 50% of the variation, followed by slope and mean annual temperature. For habitat quality, annual evapotranspiration, slope, and elevation were the dominant factors. Slope exerted a stronger impact on short-term variations in habitat quality than in water yield, with  $q$ -values exceeding 0.5. In contrast, slope orientation and annual precipitation exerted relatively minor effects on short-term fluctuations across all three ES.

In addition to identifying dominant single factors, this study examined the interactions among them. The results indicated that factor interactions provided stronger explanatory power for short-term ES fluctuations than single factors (Fig. 12). This factor-enhancing effect highlights that such fluctuations are driven by the synergistic action of multiple complex factors. Among these, the strongest two-factor enhancements involved land use inten-

Table 6 Single factor detection of short-term fluctuations

| Drivers                         | Carbon storage |         |                | Water yield |         |                | Habitat quality |         |                |
|---------------------------------|----------------|---------|----------------|-------------|---------|----------------|-----------------|---------|----------------|
|                                 | q-value        | p-value | Impact ranking | q-value     | p-value | Impact ranking | q-value         | p-value | Impact ranking |
| Slope(× 1)                      | 0.542          | 0       | 2              | 0.487       | 0       | 3              | 0.521           | 0       | 3              |
| Soil organic content(× 2)       | 0.231          | 0       | 7              | 0.320       | 0       | 7              | 0.288           | 0       | 7              |
| Elevation(× 3)                  | 0.453          | 0       | 3              | 0.404       | 0       | 6              | 0.496           | 0       | 4              |
| Aspect(× 4)                     | 0.019          | 0.023   | 9              | 0.011       | 0.204   | 9              | 0.004           | 0.620   | 9              |
| Average annual temperature(× 5) | 0.375          | 0       | 5              | 0.424       | 0       | 4              | 0.420           | 0       | 5              |
| Annual precipitation(× 6)       | 0.069          | 0       | 8              | 0.080       | 0       | 8              | 0.033           | 0       | 8              |
| Annual evapotranspiration(× 7)  | 0.422          | 0       | 4              | 0.547       | 0       | 2              | 0.524           | 0       | 2              |
| Population density(× 8)         | 0.337          | 0       | 6              | 0.411       | 0       | 5              | 0.410           | 0       | 6              |
| Land-use intensity(× 9)         | 0.900          | 0       | 1              | 0.809       | 0       | 1              | 0.947           | 0       | 1              |

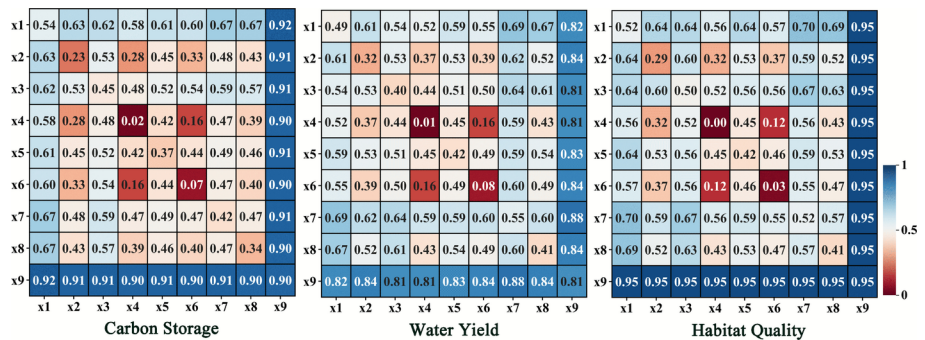


Fig. 12 The interactive detection results of various factors of short-term fluctuations

sity combined with other variables, exerting the greatest influence on all three ES. The interaction between slope and evapotranspiration also showed similarly strong explanatory power. Notably, although population density alone had only moderate explanatory power, its interactions with topographic factors such as elevation and slope exhibited strong effects. Taken together, the interaction results further suggest that land use intensity, slope, and elevation are the major drivers of short-term ES fluctuations.

## 5 Discussions

### 5.1 Ecological status assessment underlying the uncertainty of ES

In this study, the CV of carbon storage, water yield, and habitat quality were calculated for 2022–2032 in Shenzhen to assess the uncertainty of ES. Building on this, pixel-scale ES trends were analyzed and combined with baseline levels to clarify the specific ecological status underlying ES uncertainty. Unlike previous studies that assessed future ES at only a few time cross-sections, this study conducted year-by-year continuous modeling of ES for 2022–2032. The finer temporal resolution provides a critical basis for analyzing both long-term trends and short-term fluctuations of ES. It captures the continuous dynamics of ES more effectively than static snapshots.

The findings of this study are broadly consistent with the few previous studies that examined ES uncertainty, all of which noted varying degrees of uncertainty across ES and highlighted the limitations of relying solely on single-cycle snapshots for decision-making. For example, Xu et al. (2022) developed an uncertainty assessment protocol for ES-LCA based on European urban soil remediation cases, identifying the primary sources of uncertainty to mitigate major decision risks. Djahangard et al. (2024) quantified the relative contributions of different uncertainty sources to two key forest ES in Rhineland-Palatinate, Germany. Schulp et al. (2014) reviewed how uncertainty was handled in ES assessments across multiple European cities and proposed a framework for integrating uncertainty information into urban planning decisions. Some differences arise in interpretation: previous studies concluded that areas with high ES supply and low uncertainty were sustainable (Harmáčková & Vačkář, 2018), whereas areas with both high ES supply and high uncertainty could not be regarded as having favorable ecological status (Xing et al., 2020). This study offers a different perspective on these findings. We further examined the direction of change underlying ES uncertainty by analyzing ES trends for 2022–2032. We found that habitat quality and water yield exhibited high future uncertainty but were dominated by improvement with high ES. In other words, the high uncertainty of ES in these areas stemmed from continuous improvement, suggesting that ES uncertainty in high-ES regions shouldn't be arbitrarily suppressed, as it may indicate high ecological potential. This suggests that uncertainty is not inherently negative. For example, uncertainty during periods of ecological restoration can be benign, whereas urbanization-induced ecological deterioration may result in harmful uncertainty.

The five ecological statuses identified in this study were designed to guide ecological restoration by providing a comprehensive understanding of ecosystem dynamics. Among the five statuses, improvement with high ES is the most desirable, indicating that the ecosystem can deliver greater and better services to humans in the future. Areas of sustained improvement could be designated as ecological compensation priorities to maintain the upward trend in ES. Improvement with low ES partly reflected the effectiveness of ecological restoration and is therefore worth replicating. Areas showing improvement with high carbon storage were mainly concentrated in forests bordering cropland, underscoring the importance of returning cropland to forest. The conversion of cropland to built-up land had a stronger impact on improvement in high water yield.

Because degradation trends are often overlooked in areas with high ES, this ecological status is particularly noteworthy. Degradation with low ES indicated that the ecological management in these areas was either absent or ineffective, requiring urgent priority protec-



tion. It is proposed that an ecological priority restoration line be established in these areas, and that new projects incorporate green roofs and vertical greening to offset ecological losses. Unlike habitat quality, which did not exhibit low-level degradation, degradation with low carbon storage was driven by the conversion of cropland to built-up land and showed a significant trade-off with water yield. Therefore, decision-makers are advised to balance the trade-offs between the two in planning and to prioritize restoring areas of degradation with low ES.

## 5.2 Indications from ES short-term fluctuations

Short-term fluctuations of ES refer to changes occurring over relatively brief periods, which cannot be fully captured by static snapshots or long-term trend assessments. Without adequate understanding of short-term ecosystem dynamics, managers may misjudge ES supply, resulting in inefficient management. In recent years, hotspot identification has been widely applied in ecological policy and landscape planning. Short-term ES fluctuations and their spatial heterogeneity can be revealed by quantitatively analyzing annual changes in ES hotspot (Snäll et al., 2021; Srivathsa et al., 2023). In this study, we quantified interannual ES fluctuations through hotspot identification and frequency analysis. We found that carbon storage and habitat quality were dominated by low-frequency hotspot, whereas water yield was dominated by medium- to high-frequency hotspots. This indicates that an area may serve as an ES hotspot in one year but lose its high-quality ES provision in the next, highlighting the drawbacks of decision-making that relies solely on static ES assessments. Therefore, we propose a tripartite spatial zoning framework-comprising ecological core stability zones, ecological fluctuation and migration zones, and land-use conflict and synergy potential zones-that incorporates ES dynamics and volatility to provide practical guidance for Shenzhen.

The ecological core stability zones, mainly covering the mountainous areas of the Dapeng Peninsula and the contiguous forests in central-western Shenzhen (e.g., Wutong Mountain and Yangtai Mountain), serve as long-term stable providers of high-value ES. For these persistent hotspots, the primary management strategy is strict hierarchical control. They should be designated as core areas within the ecological protection red line, with all forms of development strictly prohibited to safeguard these critical ecological nodes. However, a differentiated strategy is needed for adjacent peripheral regions-such as the valleys in Biling Street (Pingshan District) and the forest-road corridors in Haishan Street (Yantian District)-which are characterized by low-frequency hotspots. In these areas, establishing flexible buffer zones is crucial. Within these buffers, large-scale urban development should be restricted, while low-impact, eco-friendly activities may be permitted. This strategy helps buffer the core areas against edge effects and preserves space for potential hotspot migration.

Close attention should also be directed to the ecological fluctuation and migration zones, including the peripheries of major reservoirs (e.g., Tiegang Reservoir in Shiyan Street, Bao'an District; Xili Reservoir in Xili Street, Nanshan District) and areas along major transportation corridors such as the Eastern Expressway in Luohu District. The marked interannual volatility of hotspots in these areas reduces the effectiveness of rigid conservation boundaries. Consequently, adopting a dynamic boundary management approach is essential. For example, areas such as the watershed of Tiegang Reservoir could be designated as hydro-ecological buffer zones, with protection boundaries subject to periodic reassessment based on annual monitoring data. In addition, to enhance resilience, constructing a distributed and modular

Green Infrastructure (GI) network is recommended along key potential migration pathways, such as the urban renewal area around Shenzhen East Railway Station in Buji Street. Such measures help maintain ecological connectivity despite spatial shifts in service provision.

Finally, addressing the land-use conflict and synergy potential zones—primarily situated in highly urbanized areas such as Yuehai Street (Nanshan District) and around major infrastructure like Bao'an Airport (Fuyong Street, Bao'an District)—requires integrated strategies to reconcile conflicting ES objectives. In these areas, which function as hotspots for water yield but coldspots for habitat quality, planning should prioritize multifunctional spatial configurations to maximize co-benefits. Specifically, in high-density urban cores such as Qianhai, regulations could encourage the adoption of elevated green roofs that simultaneously provide stormwater management and pollinator habitat, combined with subsurface infiltration structures to enhance hydrological functions while maximizing surface space efficiency. Furthermore, informed by the driver analysis of ES short-term fluctuations, explicit “sensitivity constraint indicators” should be incorporated into the planning of emerging areas. For instance, in the master plan for Guangming Science City (Guangming District), zones exceeding a defined threshold of land-use intensity could be automatically included within flexible management units.

In conclusion, future policies should move beyond static approaches by regularly monitoring regional ecological status and flexibly adjusting the boundaries of key areas in response to real-time data. This will not only support the development of a dynamically adaptive ecological management framework for Shenzhen, but also offer a transferable reference for other megacities facing fluctuations in ES.

### 5.3 Limitations and prospects

This study focuses on the dynamics of future ES by combining long-term trends and short-term fluctuations to explore the ecological status underlying ES uncertainty from a sustainability perspective. However, several limitations remain: (1) Although long-term trends and short-term fluctuations of different ES were analyzed, the dynamics interrelationships among multiple ES were not fully examined; (2) Due to the data limitations for future simulations, only nine variables were included in detecting the drivers of short-term ES fluctuations, potentially neglecting other critical factors; and (3) The short- long-term dynamics of ES were examined only for the decade 2022–2032, which constrains the temporal scope of the findings. Future studies should move beyond ES supply alone to investigate short-term fluctuations in the supply–demand balance, and incorporate three-dimensional spatial pattern indicators to better explain their drivers. They could also identify ecological tipping points for ES under multiple scenarios by coupling future climate and socioeconomic trajectories.

## 6 Conclusions

The core objective of this study was to overcome the limitations of static snapshot assessments and to reveal the long- and short-term dynamics of ES on continuous time scales. Using year-by-year simulations of Shenzhen's three key ES from 2022 to 2032, we quantified ES uncertainty, analyzed long-term trends and short-term fluctuation patterns, and explored their driving factors. The main conclusions are summarized as follows:



- 1) Regions with low carbon storage exhibited strong uncertainty, while areas of extreme habitat quality uncertainty covered the largest extent (92 km<sup>2</sup>). Water yield remained relatively stable, dominated by moderate and weak uncertainty.
- 2) Under the HDS, all three ES displayed localized patterns of improvement and degradation. Water yield and habitat quality were dominated by improvement with high ES, whereas carbon storage was primarily characterized by degradation with low ES.
- 3) All three ES showed varying degrees of short-term volatility from 2022 to 2032. Habitat quality showed the greatest fluctuation, with IFR reaching 14 km<sup>2</sup>, followed by carbon storage, while water yield remained comparatively stable. Land use intensity, slope, and elevation were identified as the main factors for short-term ES fluctuations.

The findings can support policymakers in formulating more flexible and adaptive ecological management strategies, thereby enhancing the sustainability of urban development.

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**Data availability** Data will be made available by the authors on reasonable request.

## Declarations

**Conflicts of interest** The authors declare no conflicts of interest.

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